

Digital Systems

Tutorial 3

March 2023

Tutorial:

1. Questions 1 through 6 prepared by Abhinav. G.
2. Questions 7 through 12 prepared by Shruthi. R.

Practice Problem:

1. Questions 1 through 4 prepared by Abhinav. G.
2. Questions 5 through 8 prepared by Shruthi. R.

Note: Assume both the normal and complement inputs are available

1. Truth table with input a,b,c,d and output g_3, g_2, g_1, g_0 . Implement the output g_1 using NOR gates and g_2 and g_0 using NAND gates.

Inputs	Outputs
<i>abcd</i>	<i>g₃g₂g₁g₀</i>
0000	0000
0001	0001
0010	0011
0011	0010
0100	0110
0101	0111
0110	0101
0111	0100
1000	1100
1001	1101
1010	1111
1011	1110
1100	1010
1101	1011
1110	1001
1111	1000

Ans.

	c'd'	c'd	cd	cd'
a'b'		1		1
a'b		1		1
ab		1		1
ab'		1		1

$g_0 = cd' + dc'$

	c'd'	c'd	cd	cd'
a'b'				
a'b				
ab	1	1	1	1
ab'	1	1	1	1

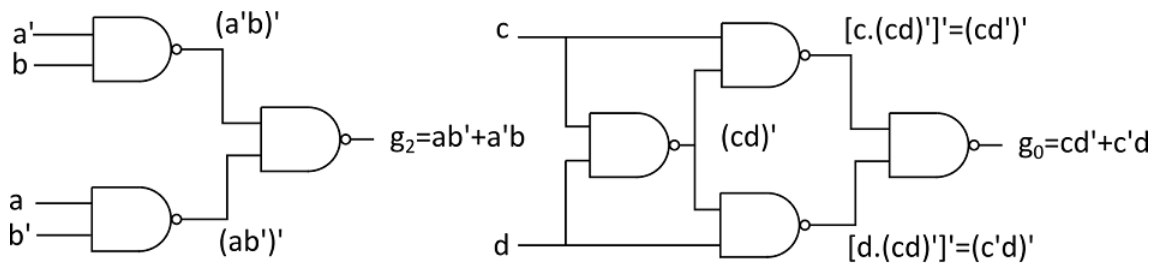
$g_3 = a$

	c'd'	c'd	cd	cd'
a'b'	0	0	1	1
a'b	1	1	0	0
ab	1	1	0	0
ab'	0	0	1	1

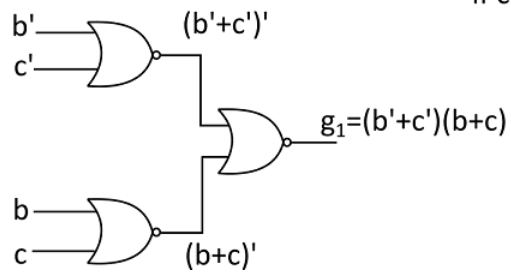
$g_1 = (b+c)(b'+c')$

	c'd'	c'd	cd	cd'
a'b'				
a'b	1	1	1	1
ab				
ab'	1	1	1	1

$g_2 = a'b + b'a$



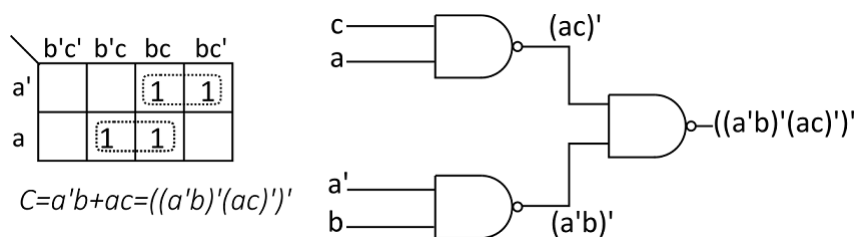
if complementary inputs not available



- Design a logic circuit for the following truth table using the minimum number of 2-input NAND gates.

x	y	z	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Ans.



3. Draw the logic diagram using only two-input NOR gates to implement the following

$$F(A, B, C, D) = (A \oplus B)'(C \oplus D)$$

Ans. $F = (AB + A'B')(CD' + C'D)$

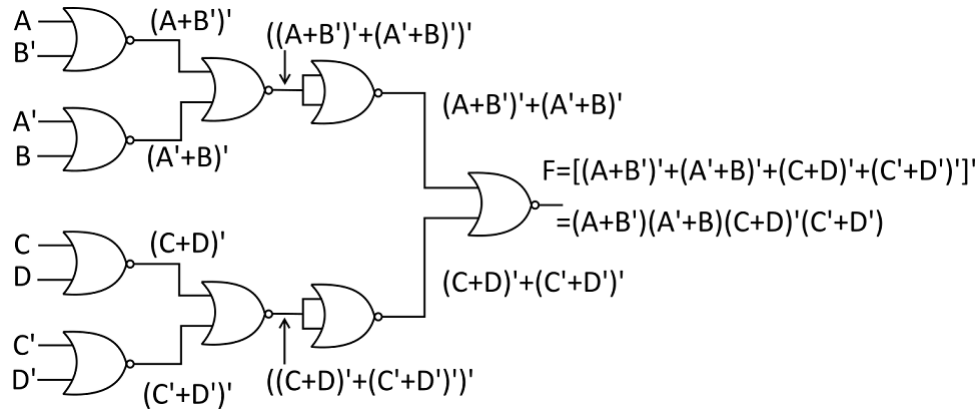
$$F = ABCD' + ABC'D + A'B'CD' + A'B'C'D$$

	C+D	C+D'	C'+D'	C'+D
A+B	0	1	0	1
A+B'	0	0	0	0
A'+B'	0	1	0	1
A'+B	0	0	0	0

$$F = (A + B')(A' + B)(C + D)(C' + D')$$

$$F = [((A + B')(A' + B))' + ((C + D)(C' + D'))']'$$

$$F = [(((A + B')' + (A' + B)')')' + (((C + D)' + (C' + D')')')']'$$



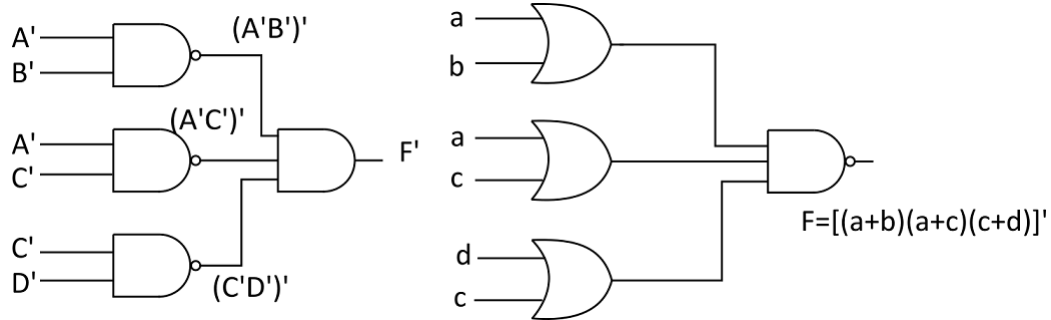
4. Design a logic circuit for F using OR-NAND and for F' using NAND-AND for the two-level forms for the following expression.

$$F = [(A + B)(A + C)(C + D)]'$$

Ans. $F = [(A + B)(A + C)(C + D)]'$

$$F = [(A'B')'.(A'C')'.(C'D')']'$$

$$F' = (A'B')'.(A'C')'.(C'D')$$



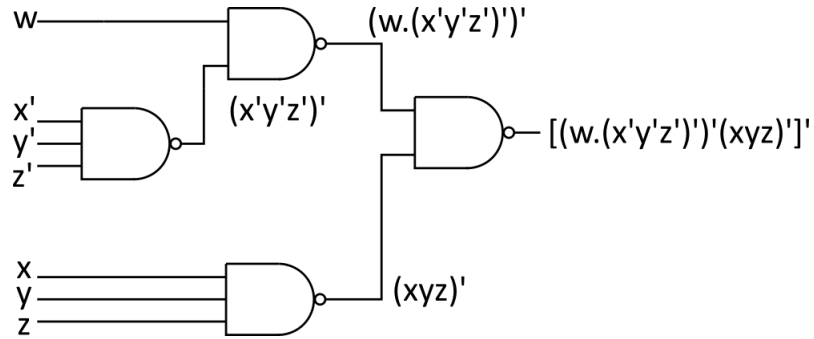
5. Show that the dual of the exclusive-OR is also its complement.

Ans. $x \oplus y = \bar{x}y + x\bar{y}$ Dual: $(\bar{x} + y).(x + \bar{y}) = \bar{x}.\bar{y} + xy = \overline{x \oplus y}$

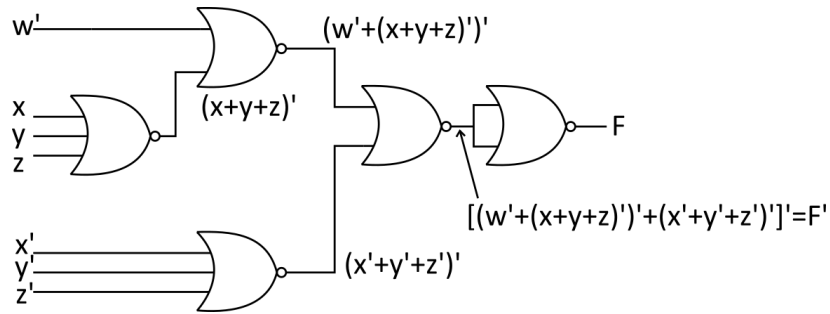
6. Draw the multiple-level circuit for the following equation using:
(i) NAND (b) NOR

$$w(x + y + z) + xyz$$

Ans. (i) $F = [(w(x + y + z))'.(xyz)']' = [(w(x'y'z'))'.(xyz)']'$



$$(ii) F' = [w(x + y + z) + xyz]' = [(w' + (x + y + z)')' + (x' + y' + z')']'$$



7. Design a logic circuit using NAND gates that has three inputs, A, B, and C, and whose output will be HIGH only when a majority of the inputs are HIGH.

Solution:

Step 1. Set up the truth table and write down the minterms:

On the basis of the problem statement, the output x should be 1 whenever two or more inputs are 1 and for all other cases, the output should be 0.

A	B	C	x	
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	1	$\rightarrow \bar{A}BC$
1	0	0	0	
1	0	1	1	$\rightarrow A\bar{B}C$
1	1	0	1	$\rightarrow AB\bar{C}$
1	1	1	1	$\rightarrow ABC$

$$x(A, B, C) = \sum(3, 5, 6, 7)$$

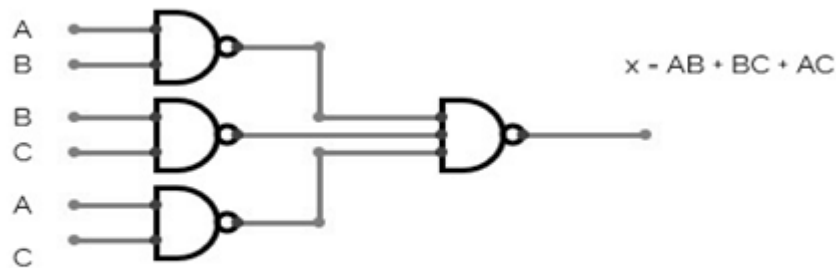
Step 2. Solve the above truth table using K-map:

$A \backslash B, C$	00	01	11	10
0	0 0	0 1	1 3	0 2
1	0 4	1 5	1 7	1 6

Step 3. Write the sum-of-products expression for the output.

$$x = BC + AC + AB$$

Step 4. Implement the circuit for the final expression



8. Implement the following Boolean function $F(A, B, C, D) = \sum(0, 1, 2, 9, 11)$ together with the don't-care condition $d(A, B, C, D) = \sum(8, 10, 14, 15)$ using no more than two NOR gates. Assume that both the normal and complement inputs are available.

Solution:

Step 1: Plot the K-map for the Boolean function:

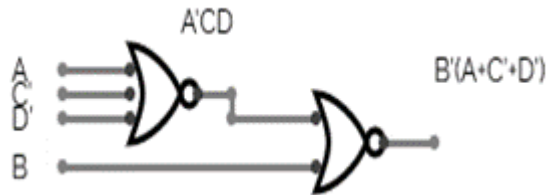
A,B\ C,D	00	01	11	10
00	1 0	1 1	0 3	1 2
01	0 4	0 5	0 7	0 6
11	0 12	0 13	- 15	- 14
10	- 8	1 9	1 11	- 10

Step 2: Write the POS expression for the output:

$$F' = B + A'CD$$

$$F = B'(A + C' + D')$$

Step 3: Implement the circuit using only two NOR gate:



9. Design a logic circuit for each output using NAND gates with three inputs, x, y and z, and the three outputs, A, B, and C. when the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When the binary input is 4, 5, 6, or 7, the binary output is one less than the input.

Solution:

Design procedure:

- Derive the truth table that defines the required relationship between inputs and outputs.

x	y	z	A	B	C
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	0	1	1
1	0	1	1	0	0
1	1	0	1	0	1
1	1	1	1	1	0

- Obtain the simplified Boolean functions for each output as a function of the input variables.

K-Map for output A: The simplified expression from the k-map is:

		yz				y	
x		00	01	11	10		
0		0	0	1	0		
1		0	1	1	1		

z

$$A = xy + xz + yz$$

K-Map for output B: The simplified expression from the k-map is:

		yz				y	
x		00	01	11	10		
0		0	1	0	1		
1		1	0	1	0		

z

$$B = x'y'z + x'yz' + xy'z' + xyz$$

K-Map for output C: The simplified expression from the k-map is:

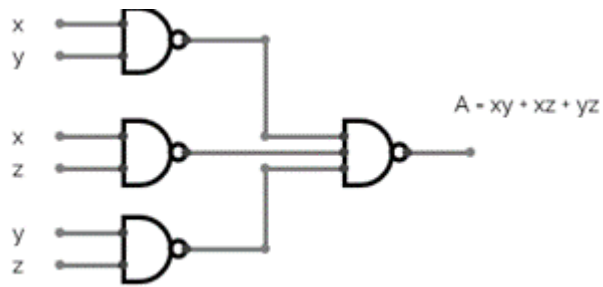
		yz				y	
x		00	01	11	10		
0		1	0	0	1		
1		1	0	0	1		

z

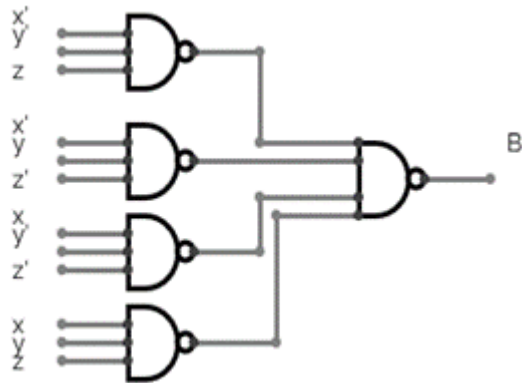
$$C = z'$$

3. Draw the logic diagram.

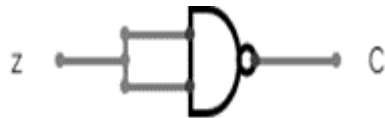
$$A = xy + xz + yz$$



$$B = x'y'z + x'yz' + xy'z' + xyz$$



$$C = z'$$



10. A car alarm system is to be designed considering 4 inputs door closed(D),key in(K),seat pressure(S), seat belt closed(B).

The alarm A should sound if: 1. the key is in and door is not closed 2. the door is closed the key is in, driver in the seat and seat belt is not closed

The system is to be designed with 2 input NOR gates and inputs are available in basic forms only. How many number of gates are required???

Solution:

1-door is closed else it is open

1-key is in

1-driver in the seat

1-seat belt is closed

Draw truth table for 4 inputs

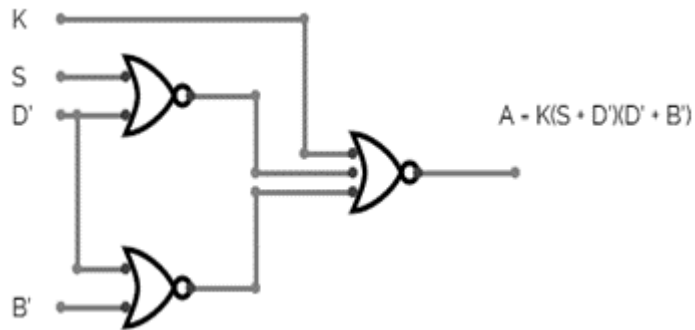
D	K	S	B	A
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0

If we simplify using K-map

$$A(D, K, S, B) = \prod(0, 1, 2, 3, 8, 9, 10, 11, 12, 13, 15)$$

$D, K \backslash S, B$	00	01	11	10
00	0 0	0 1	0 3	0 2
01	1 4	1 5	1 7	1 6
11	0 12	0 13	0 15	1 14
10	0 8	0 9	0 11	0 10

POS is $A = K(D' + S)(D' + B')$

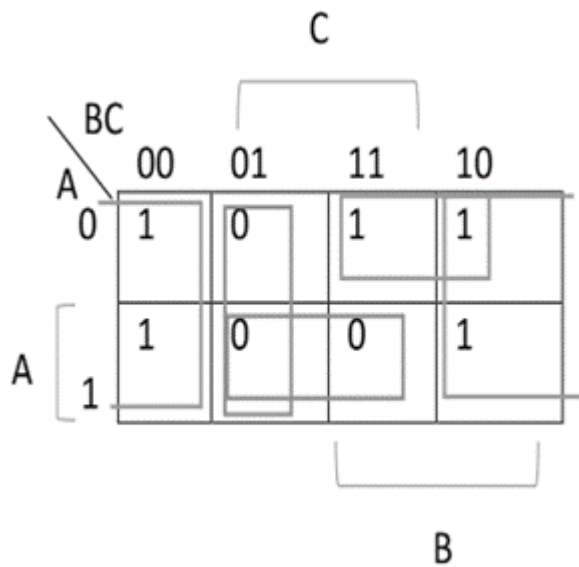


Hence, total 3 NOR gates are required.

11. Implement the function $F(A, B, C) = \sum(0, 2, 3, 4, 6)$ using 2-level AND-NOR, NAND-AND, OR-NAND and NOR-OR logic gates after simplifying it in
 (a) Sum of products. b) Product of sums:

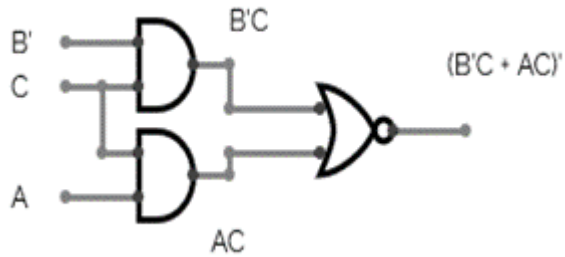
Solution:

Solve the K-map for the function $F(A, B, C) = \sum(0, 2, 3, 4, 6)$



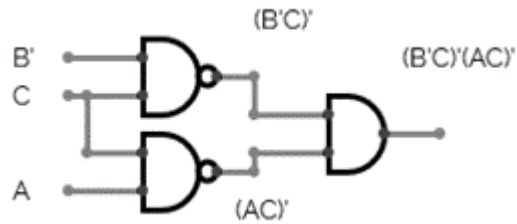
- (a) For AND-NOR and NAND-AND implementation of the function. Consider the SOP form by combining 0's in the K-map:
 $F' = B'C + AC$

$$F = (B'C + AC)' = (B + C')(A' + C')$$



AND-NOR

$$(B'C + AC)' = (B + C')(A' + C')$$



NAND-AND

$$(B'C)'(AC)' = (B + C')(A' + C')$$

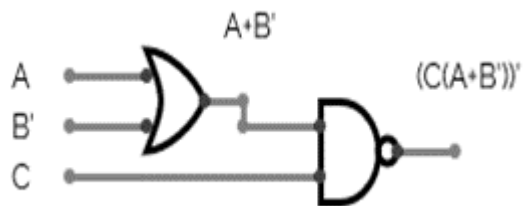
(b) For OR-NAND and NOR-OR implementation of the function. Consider the POS form by combining 1's in the K-map and complement it:

$$F = C' + A'B$$

Taking Complement:

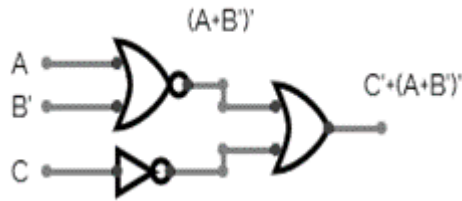
$$F' = C(A + B')$$

$$F = (C(A + B'))' = C' + (A'B)$$



OR-NAND

$$(C(A + B'))' = C' + (A'B)$$



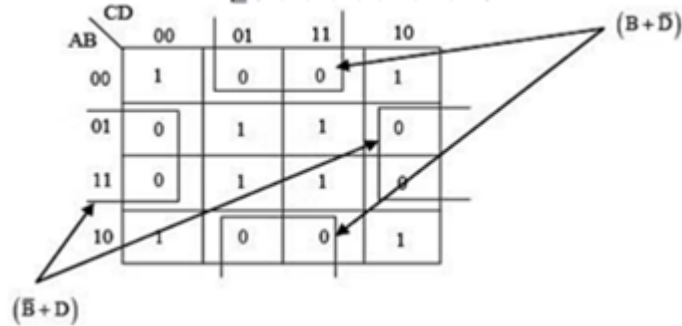
NOR-OR

$$C' + (A + B')' = C' + (A'B)$$

12. What is the minimum number of 2-input NOR gates required to implement a 4- variable function expressed in sum-of-minterms form as $F(A,B, C,D) = \sum(0, 2, 5, 7, 8, 10, 13, 15)$. Assume that all the inputs and their complements are available.

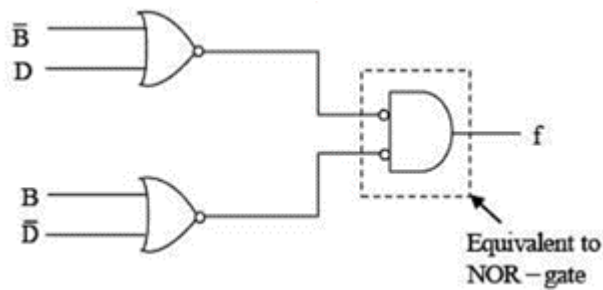
Solution: To implement the 4-variable function using 2-input NOR gate only, first convert the SOP form of function into POS form and then solve the K-map.

$$\text{POS form: } F(A,B, C,D) = \prod(1, 3, 4, 6, 9, 11, 12, 14)$$

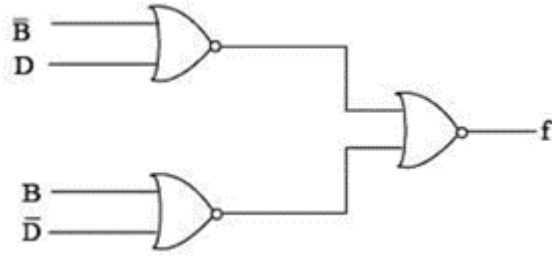


$$f = (\overline{B} + D)(B + \overline{D})$$

As per the questions their complemented and normal both type of inputs are available.



Their equivalent circuit can be redrawn as



Therefore, minimum number of NOR-gate = 3.

Practice Problems

1. The logic circuit with 4 inputs a, b, c, d and 5 outputs p, q, r, s, t. When binary inputs are greater than 7, then binary output is two greater than the input otherwise input is equal to output. Draw the circuit using NAND gates for output r and NOR gates for output q. Can you guess which converter is this?

Ans. It is a decimal to octal converter.

$t = d$ just by observation.

abcd	pqrst
0000	00000
0001	00001
0010	00010
0011	00011
0100	00100
0101	00101
0110	00110
0111	00111
1000	01010
1001	01011
1010	01100
1011	01101
1100	01110
1101	01111
1110	10000
1111	10001

	c+d	c+d'c'+d'c'+d		
a+b	0	0	0	0
a+b'	0	0	0	0
a'+b'	1	1	0	0
a'+b	1	1	1	1

$q = a(b' + c')$

	c'd'	c'd	cd	cd'
a'b'				
a'b	1	1	1	1
ab	1	1		
ab'			1	1

$r = a'b + bc' + ab'c$

	c'd'	c'd	cd	cd'
a'b'				
a'b				
ab			1	1
ab'				

$p = abc$

	c'd'	c'd	cd	cd'
a'b'			1	1
a'b			1	1
ab	1	1		
ab'	1	1		

$s = a'c + a'c'$

q is written in POS:

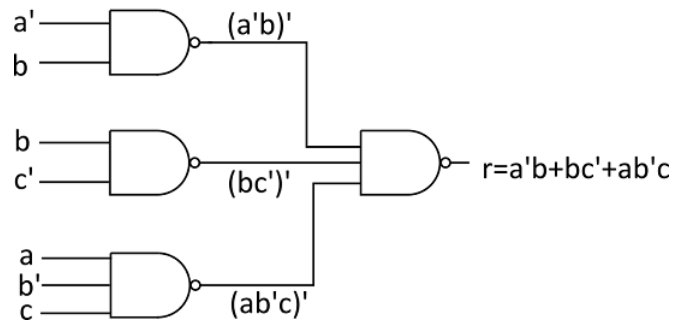
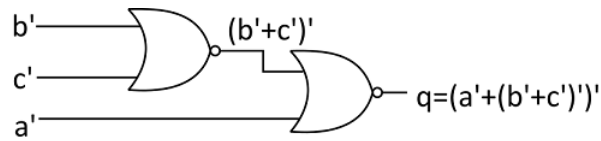
$$q = a(b' + c')$$

$$q = [a' + (b' + c')']'$$

r is written in SOP:

$$r = a'b + bc' + ab'c$$

$$r = ((a'b)'(bc')'(ab'c)')'$$



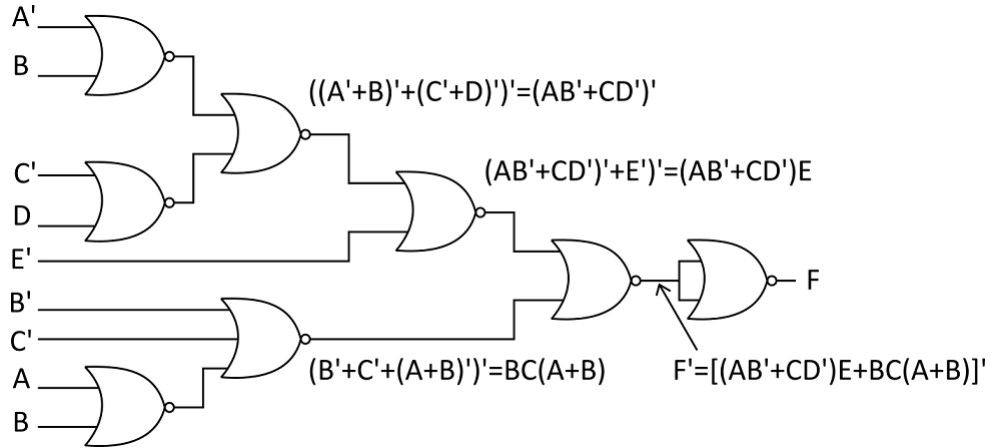
2. Draw multiple-level NOR circuit for the following expression:

$$(AB' + CD')E + BC(A + B)$$

Ans. $F = (AB' + CD')E + BC(A + B) = [(AB' + CD')' + E']' + [(BC)' + (A + B)']'$

Replacing AB' by $(A' + B)'$, CD' by $(C' + D)'$

$$F = [((A' + B)' + (C' + D)')' + E']' + [(B' + C') + (A + B)']'$$



3. Derive the circuits for a three-bit parity generator and a four-bit parity checker using an odd-parity bit.

Ans.

ABC		ABCP		E	
000	1	0000	1	0000	1
001	0	0001	0	0001	0
010	0	0010	0	0010	0
011	1	0011	1	0011	1
100	0	0100	0	0100	0
101	1	0101	1	0101	1
110	1	0110	1	0110	1
111	0	0111	0	0111	0
				1000	0
				1001	1
				1010	1
				1011	0
				1100	1
				1101	0
				1110	0
				1111	1

	B'C'	B'C	BC	BC'
A'	1		1	
A		1		1

	C'P'	C'P	CP	CP'
A'B'	1		1	
A'B		1		1
AB	1		1	
AB'		1		1

$$P = A'B'C' + AB'C + A'BC + ABC' = A(BC' + B'C) + A'(B'C' + BC)$$

$$P = A(B \oplus C) + A'(B \oplus C)' = A \odot (B \oplus C)$$

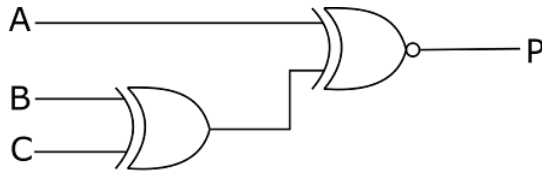
$$E = A'B'C'P' + A'B'CP + A'BC'P + A'BCP' + ABC'P' + ABCP + AB'C'P + AB'CP'$$

$$E = A'B'(C'P' + CP) + A'B(C'P + CP') + AB(C'P' + CP) + AB'(C'P + CP')$$

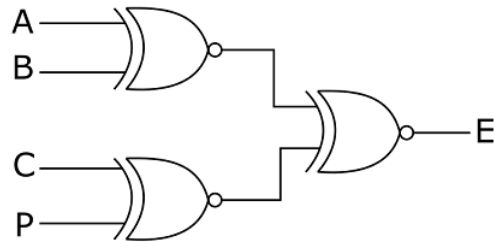
$$E = (AB' + A'B)(C'P + CP') + (A'B' + AB)(C'P' + CP)$$

$$E = (A \oplus B)(C \oplus P) + (A \odot B)(C \odot P) = (A \odot B)(C \odot P) + (A \odot B)'(C \odot P)'$$

$$E = (A \odot B) \odot (C \odot P)$$



parity generator



parity checker

4. Draw the NOR circuit for the following truth table where A , B and B_{in} are the inputs and B_{out} is the output.

A	B	B_{in}	B_{out}
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

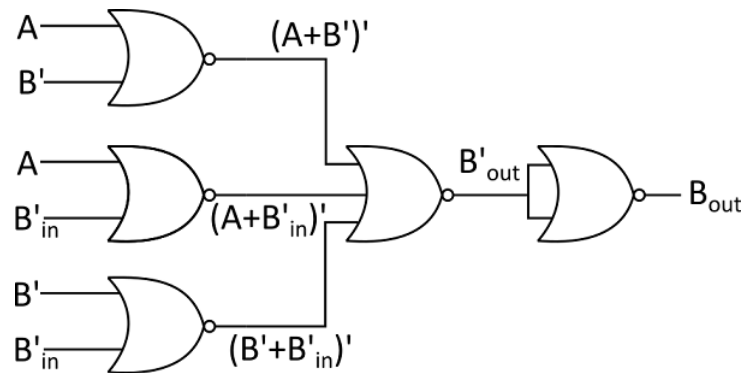
Ans.

	$B'B_{in}$	$B'B_{in}$	BB_{in}	BB_{in}
A'	1	1	1	
A		1		

$$B_{out} = A'B + A'B_{in} + BB_{in}$$

$$B_{out} = (A + B')' + (A + B'_{in})' + (B' + B'_{in})'$$

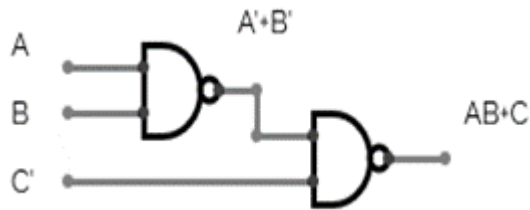
$$B'_{out} = [(A + B')' + (A + B'_{in})' + (B' + B'_{in})']'$$



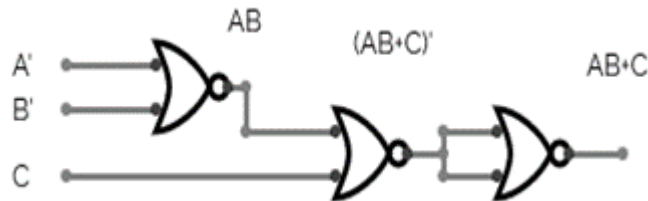
5. How many minimum 2-inputs NAND and NOR gate will be required to implement: $AB + C$

Solution:

2 minimum NAND gate are required to implement the given function if input complement are available else 3 minimum NAND gate are required.



3 minimum NOR gate are required to implement the given function if input complement are available else 5 minimum NOR gate are required.



6. Design a logic circuit using NAND gate for each b and c output that accepts a 3 bit number as input and generate an output binary number equal to square of the input number. (consider: x,y,z as inputs and a,b,c,d,e,f as outputs)

Solution:

Largest 3 bit number is $= (111)_2 = (7)_{10}$

So square of $(7)^2 = (49)_{10} = (110001)_2$ that is ,it requires 6 bits to represent output.

Truth Table :

Inputs			Outputs					
X	Y	Z	a	b	c	d	e	f
0	0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0	1
0	1	0	0	0	0	1	0	0
0	1	1	0	0	1	0	0	1
1	0	0	0	1	0	0	0	0
1	0	1	0	1	1	0	0	1
1	1	0	1	0	0	1	0	0
1	1	1	1	1	0	0	0	1

solving for above outputs,

We see that, the all output of e is equal to zero and the all output of f is equal to the input z. so, $e = 0$ and $f = z$

Now we draw the map for a,b,c,d

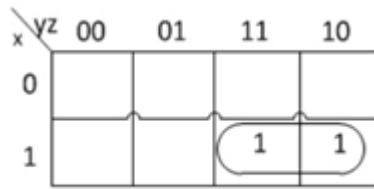


Fig: Map for a
 $a = xy$

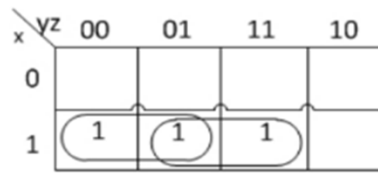


Fig: Map for b
 $b = xy' + xz$

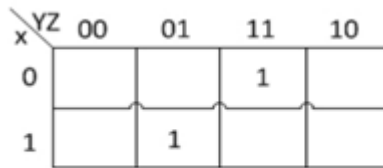


Fig: Map for c
 $C = xy'z + x'yz$
 $c = z(x \oplus y)$

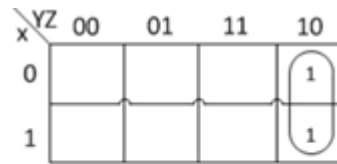
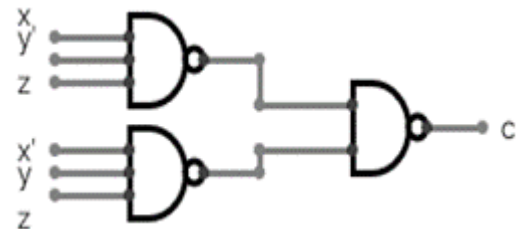
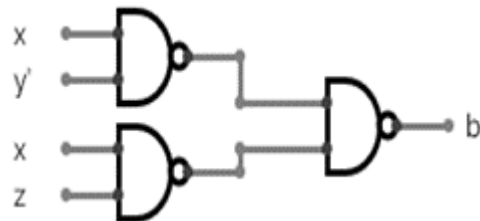
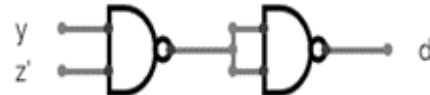
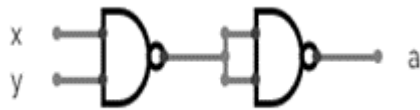


Fig: Map for d
 $d = yz'$

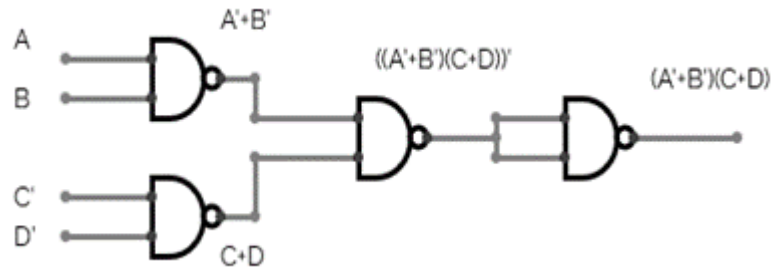
Now the logic circuit will be,



7. How many minimum numbers of NAND gate require to implement expression: $(\overline{A+B})(C+D)$

Solution:

If we consider complement of inputs are available then 4 minimum NAND gate required to implement the expression



If complement inputs are not available then 6 minimum NAND gate required to implement expression

8. Give the NAND implementation of a circuit, using minimum gates, that gives HIGH when the input is BCD equivalent of 5, 7 or 9 and LOW otherwise?

Solution:

Step 1. Set up the truth table: As the input is a BCD number, the output will be don't care for the input combinations, 10, 11, 12, 13, 14 and 15.

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Step 2. So the K-Map will be as shown below:

CD \ AB		CD			
		00	01	11	10
AB	00				
	01		1	1	
	11	X	X	X	X
	10		1	X	X

Step 3. The simplified Boolean Function: $Y = AD + BD$

Step 4. Implement the circuit for the final expression

